Life Expectancy Across Nations: Patterns and Predictors

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1. Introduction

The life expectancy at birth is a vital indicator of the health of the population, representing the total well-being of the population, health care delivery facilities, and the socio-economic situation in a given country. Knowing what pushes the disparities in life expectancy between countries will enable policymakers to develop specific measures that will enhance their health results. The paper will provide a cross-country analysis of variation of life expectancy based on a dataset of 42 countries of choice as of 2018. The year 2018 was selected because it is the year that has all complete and consistent data in all countries, preventing the problem of other years that may cause cross-sectional analysis to be difficult(Cutler, Deaton, & Lleras-Muney, 2006). The most important explanatory variables are log GDP per capita (PPP), current health expenditure per capita, total fertility rate, and availability of electricity. These variables are economic, demographic, and infrastructural factors in the health of a population.

2. Part 1: The Data Collection and Preparation

2.1 Sample Selection

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```
> dim(data_42_final)
[1] 42  6
> data_42_final$country
 [1] "Bangladesh"           "Bhutan"               "Denmark"              "India"                "Japan"
 [6] "Malaysia"             "Nepal"                "Nigeria"              "Philippines"          "Qatar"
[11] "Sri Lanka"            "Sweden"               "Panama"               "Turkmenistan"         "Slovak Republic"
[16] "Azerbaijan"           "Suriname"             "Samoa"                "Dominica"             "Libya"
[21] "Congo, Rep."          "West Bank and Gaza"   "Uruguay"              "OECD members"         "IDA & IBRD total"
[26] "IDA blend"            "Syrian Arab Republic" "Uganda"               "South Asia"           "IDA only"
[31] "Middle income"        "Italy"                "Gabon"                "Brunei Darussalam"    "Angola"
[36] "Vanuatu"              "Trinidad and Tobago"  "Portugal"             "Grenada"              "Guinea-Bissau"
[41] "Upper middle income"  "Kenya"
>
```

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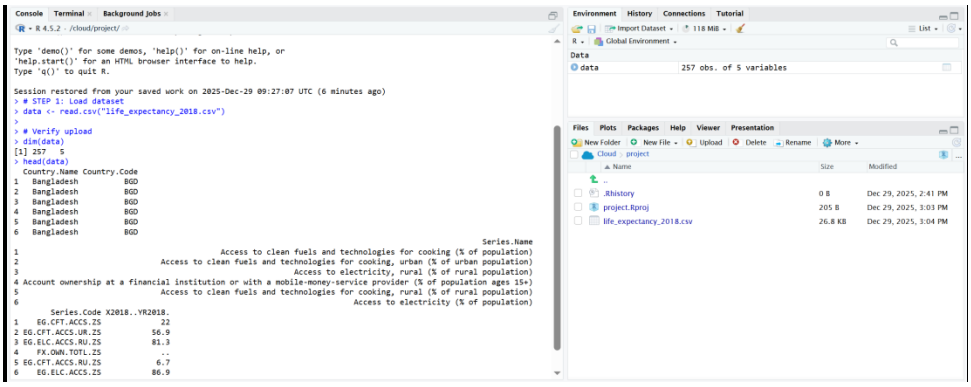

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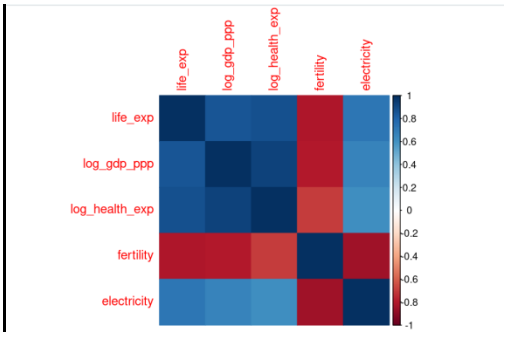

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```
> #  Clean World Bank 2018 data
> > # Remove aggregates using ISO codes
> data_clean <- data %>%
+     filter(!is.na(iso2c)) %>%
+     select(country, life_exp, gdp_ppp, health_exp, fertility, electricity) %>%
+     filter(!is.na(life_exp))
> > # Check dataset
> dim(data_clean)
[1] 265   6
> head(data_clean)
                    country life_exp   gdp_ppp health_exp fertility electricity
1                 Afghanistan 62.44300  2432.277   71.22509  5.327000    93.40000
2 Africa Eastern and Southern 63.33069  3723.216   90.07565  4.521443    43.03507
3  Africa Western and Central 57.03698  5179.360   56.12426  4.962571    51.33583
4                   Albania 79.23800 15054.998  349.21127  1.415000   100.00000
5                   Algeria 75.55500 13727.127  271.53778  3.019000    99.60000
```

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```
R - R 4.5.2 · /cloud/project/
> data_42_final$country
 [1] "Bangladesh"            "Bhutan"                "Denmark"               "India"                 "Japan"
 [6] "Malaysia"              "Nepal"                 "Nigeria"               "Philippines"           "Qatar"
[11] "Sri Lanka"             "Sweden"                "Panama"                "Turkmenistan"          "Slovak Republic"
[16] "Azerbaijan"            "Suriname"              "Samoa"                 "Dominica"              "Libya"
[21] "Congo, Rep."           "West Bank and Gaza"    "Uruguay"               "OECD members"          "IDA & IBRD total"
[26] "IDA blend"             "Syrian Arab Republic"  "Uganda"                "South Asia"            "IDA only"
[31] "Middle income"         "Italy"                 "Gabon"                 "Brunei Darussalam"     "Angola"
[36] "Vanuatu"               "Trinidad and Tobago"   "Portugal"              "Grenada"               "Guinea-Bissau"
[41] "Upper middle income"   "Kenya"
>
> #  Recalculate log variables for OLS
> data_42_final <- data_42_final %>%
+     mutate(
+         log_gdp_ppp = log(gdp_ppp),
+         log_health_exp = log(health_exp)
+     )
>
> #  Summary statistics
> summary_stats <- data_42_final %>%
+     summarise(
+         life_exp_mean = mean(life_exp),
+         life_exp_sd = sd(life_exp),
+         gdp_mean = mean(log_gdp_ppp),
+         gdp_sd = sd(log_gdp_ppp),
+         health_mean = mean(log_health_exp),
+         health_sd = sd(log_health_exp),
+         fert_mean = mean(fertility),
+         fert_sd = sd(fertility),
+         elec_mean = mean(electricity),
+         elec_sd = sd(electricity)
+     )
> summary_stats
  life_exp_mean life_exp_sd gdp_mean    gdp_sd health_mean health_sd fert_mean  fert_sd elec_mean  elec_sd
1      72.18266     6.812008 9.472338 0.9835042    5.707139  1.558567  2.622178 1.166657  86.68068 20.03157
>
```

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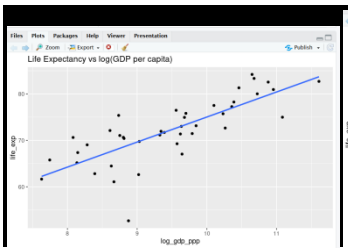

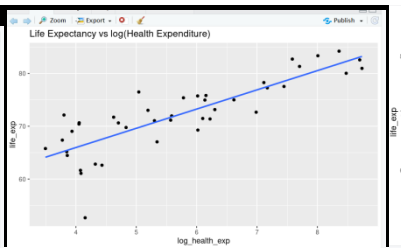

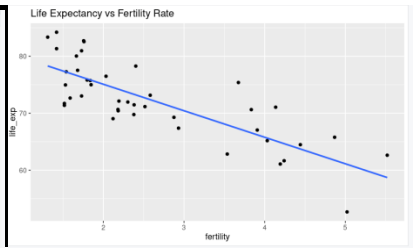

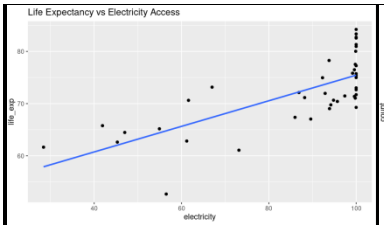

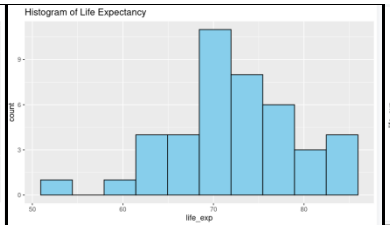

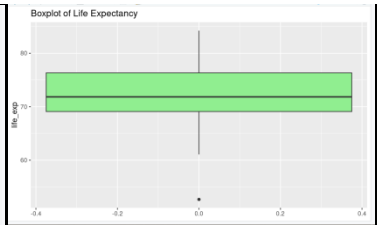

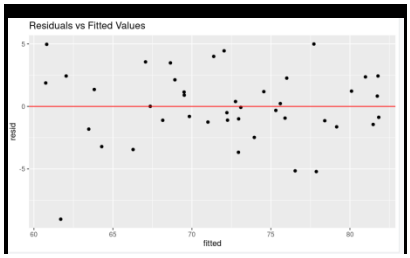

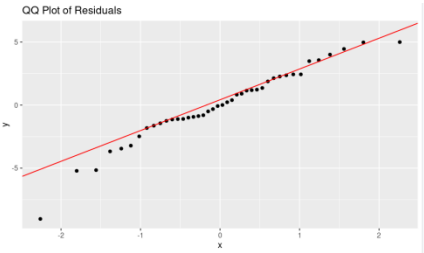

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```
> #  OLS Regression for final 42-country dataset
>
> # OLS model
> ols_model_final <- lm(
+     life_exp ~ log_gdp_ppp + log_health_exp + fertility + electricity,
+     data = data_42_final
+ )
>
> # Summary of OLS
> summary(ols_model_final)

Call:
lm(formula = life_exp ~ log_gdp_ppp + log_health_exp + fertility +
    electricity, data = data_42_final)

Residuals:
   Min     1Q Median     3Q    Max
-9.026 -1.225 -0.035  2.066  4.994

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    68.66602    9.53709   7.200 1.54e-08 ***
log_gdp_ppp    -1.11169    1.20807  -0.920 0.363421
log_health_exp  2.87792    0.71053   4.050 0.000251 ***
fertility      -2.26618    0.74994  -3.022 0.004541 **
electricity     0.04112    0.04213   0.976 0.335320
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.044 on 37 degrees of freedom
Multiple R-squared:  0.8199,	Adjusted R-squared:  0.8004
F-statistic:  42.1 on 4 and 37 DF,  p-value: 2.739e-13
```

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```
> coeftest(ols_model_final, vcov = vcovHC(ols_model_final, type = "HC1"))

t test of coefficients:

                Estimate Std. Error t value  Pr(>|t|)
(Intercept)    68.666024  11.430691  6.0072  6.14e-07 ***
log_gdp_ppp    -1.111686   1.493309 -0.7444 0.4613095
log_health_exp  2.877915   0.710152  4.0525 0.0002497 ***
fertility      -2.266177   0.873951 -2.5930 0.0135466 *
electricity     0.041123   0.038849  1.0585 0.2966795
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

>
> #  Overall F-test
> waldtest(ols_model_final)
Wald test

Model 1: life_exp ~ log_gdp_ppp + log_health_exp + fertility + electricity
Model 2: life_exp ~ 1
  Res.Df Df      F    Pr(>F)
1     37
2     41 -4 42.098 2.739e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>
>
```

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